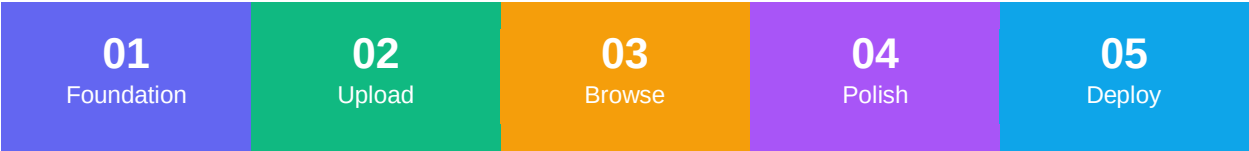


Lab Screenshot Manager

Phase-by-Phase Build Plan

A complete UI-first development roadmap across 5 focused phases



Total Estimated Duration: 3–5 weeks · Stack: React, CSS Variables, window.storage API

PHASE 01

Foundation

Design system, app shell, folder sidebar

Duration	3–4 days
Goal	Build the visual identity and empty app skeleton so every future phase has a solid design base.
Stack	React (functional components), inline CSS with CSS custom properties, Google Fonts
Outcome	A navigable shell with folders in the sidebar — no real data yet, fully styled.

Tasks

#	Task	What to build / design	Effort
1	Design Token Setup	Define all CSS variables: --color-bg, --color-surface, --color-accent, --color-text-*, --radius-*, --shadow-*. Build a tokens.css file or inline :root block.	2h
2	Font & Typography	Import a display font (e.g. 'DM Mono' or 'Space Grotesk') + a body font. Define type scale: xs/sm/base/lg/xl/2xl. Apply globally.	1h
3	App Shell Layout	Two-column flex layout: fixed 240px sidebar (dark) + flexible main area. Header bar with app name + icon. Responsive breakpoint at 768px.	3h
4	Folder Sidebar	List of folder items with active state highlight, folder icon, name, file count badge. 'New Folder' button at bottom. Empty state illustration when no folders.	3h
5	Folder Creation Modal	Click 'New Folder' -> animated slide-up modal with text input, confirm + cancel. Keyboard: Enter confirms, Esc cancels. Inline validation (empty name).	2h
6	Folder Deletion	Right-click or kebab menu on folder -> 'Delete' option -> confirmation toast/modal. Smooth collapse animation on removal.	2h
7	Empty Main State	When no folder selected or folder is empty: centred illustration + heading + subtitle. Distinct states for (a) no folders at all, (b) folder exists but empty.	1h

UI Specifications

UI Component	Design Specification
Sidebar	Dark navy (#0F172A), 240px fixed, folder rows 44px tall, 8px border-radius on active row, indigo (#6366F1) left border on active.
Active Folder	Indigo background tint (rgba(99,102,241,0.15)), white text, 3px left accent bar.
New Folder Btn	Dashed border button, indigo accent, full sidebar width, 40px height. Hover: solid border + light tint.
App Header	20px tall logo text in display font, monospaced. Right side: version badge. Border-bottom separator.
Typography	Headings: DM Mono or JetBrains Mono. Body: Inter or Outfit. Sizes: 11px–28px scale.
Colour Palette	BG #0A0A0F, Surface #111118, Border #1E1E2E, Accent #6366F1, Text-primary #F1F5F9, Text-muted #94A3B8.

Phase Outcome: A beautiful, navigable shell. You can create and switch between folders. Design system is locked in — all future phases inherit these tokens.

PHASE 02

Upload & Storage

Drag-and-drop upload, Base64 encoding, persistent storage

Duration	3–4 days
Goal	Make the app actually store files. Users can drag screenshots in and they persist across page reloads.
Stack	FileReader API, window.storage (key-value), React state + useEffect
Outcome	Files uploaded to a folder survive browser close. File count shown in sidebar badge.

Tasks

#	Task	What to build / design	Effort
1	Drop Zone Component	Full main-panel drop zone when folder is open. Dashed animated border. Visual highlight on drag-over. Click-to-open file picker fallback. Accept all file types.	3h
2	FileReader Integration	Read each dropped file as Base64 data URL via FileReader.readAsDataURL. Show per-file progress bar. Handle multiple files in parallel.	2h
3	Storage Schema	Design two storage keys: lab:folders (array) and lab:files (object keyed by folder ID). Each file entry: { id, name, type, size, data, added }.	1h
4	Persist on Upload	After reading, write to window.storage. Update React state. Handle storage errors gracefully (quota exceeded: show warning, skip file, keep others).	2h
5	Load on Mount	In useEffect, read both storage keys on app init. Reconstruct state. Show loading spinner while async reads complete.	2h
6	File Size Guard	Before encoding, check file.size. If > 4MB: show inline warning banner with file name. Do not upload. Allow user to dismiss.	1h
7	Upload Feedback	Uploading spinner overlay on drop zone. Per-file toast notifications: success (green) / skip (yellow) / error (red). Auto-dismiss after 3s.	2h

#	Task	What to build / design	Effort
8	Sidebar Badge Update	After every upload, recompute folder file count and update the badge in the sidebar instantly.	1h

UI Specifications

UI Component	Design Specification
Drop Zone (idle)	Dashed 2px border (#10B981 tint), 'Drop files here or click to browse' caption, upload icon centred, 16px border-radius.
Drop Zone (active)	Solid emerald border, green glow box-shadow, background tint rgba(16,185,129,0.08), scale(1.01) transform.
Progress Bar	Thin 3px bar below file name, emerald fill, smooth width transition, disappears after 500ms on completion.
Toast Notification	Bottom-right stack, slide-in from right, coloured left border (green/yellow/red), icon + filename + status text, 3s auto-dismiss.
Size Warning Banner	Full-width amber banner inside drop zone, warning icon, file name in bold, 'File too large (max 4MB)' message, X dismiss.
Loading State	Full-panel skeleton shimmer on initial load, 3 ghost file cards, animated pulse.

Phase Outcome: The app now stores real files. Drop screenshots in — close the tab — open again — they are still there.

PHASE 03

Display & Browse

File grid, image thumbnails, lightbox, download & delete

Duration	4–5 days
Goal	Display uploaded files beautifully. Images as thumbnails in a grid, other files as type-icon cards. Full lightbox viewer.
Stack	React state, CSS Grid, DOM URL API for blob previews
Outcome	Users can visually browse, preview full-size, download, and delete files.

Tasks

#	Task	What to build / design	Effort
1	File Grid Layout	CSS Grid with auto-fill columns (minmax(180px, 1fr)). Gap 16px. Cards with hover lift effect (translateY(-2px) + shadow). Grid → list view toggle.	3h
2	Image Thumbnail Card	Render Base64 img as card background (object-fit: cover, aspect-ratio 4/3). Overlay on hover: file name, size, date, action buttons (view, download, delete).	3h
3	Non-image File Card	Detect type from file.type. Render appropriate icon (PDF=red, TXT=gray, ZIP=yellow, CODE=cyan, etc.). Name truncated with ellipsis. Size + date below.	2h
4	Lightbox Viewer	Click thumbnail -> full-screen overlay (backdrop-filter: blur(20px)). Image centred, max 90vw/90vh. Navigation arrows if multiple images. Esc / click-outside to close.	4h
5	Keyboard Navigation	Arrow keys navigate between images in lightbox. Esc closes. Tab loops through lightbox controls. ARIA labels for screen reader support.	2h
6	Download Action	Create object URL from Base64 -> programmatic anchor click -> revoke URL. Works for all file types.	1h
7	Delete with Confirm	Trash icon on hover. Click -> inline confirmation tooltip (not modal): 'Delete? Undo / Confirm'. Confirm removes from storage + state with fade-out animation.	2h

#	Task	What to build / design	Effort
8	Sort & Filter Bar	Top bar above grid: sort by (newest, oldest, name, size) dropdown. Filter by type (all / images / documents / other) pill buttons.	3h
9	Search	Search input in top bar. Real-time filter of file names across the active folder. Highlight matches in file name.	2h

UI Specifications

UI Component	Design Specification
File Card	Background #111118, border 1px #1E1E2E, border-radius 12px, overflow hidden. Hover: border-color amber, box-shadow 0 8px 24px rgba(0,0,0,0.4).
Thumbnail Overlay	Gradient from transparent to rgba(0,0,0,0.85) bottom 60%. Action buttons appear on hover with opacity transition (0 -> 1, 200ms ease).
Lightbox Backdrop	rgba(0,0,0,0.92) + backdrop-filter blur(12px). Image: max-width 90vw, max-height 90vh, object-fit contain, border-radius 8px.
Nav Arrows	40px circle buttons, semi-transparent white bg, absolute positioned. Slide in on hover. Disabled state when at first/last image.
Type Icons	24px SVG icons. PDF: #EF4444, Word: #3B82F6, Excel: #10B981, ZIP: #F59E0B, Code: #06B6D4, Unknown: #94A3B8.
Sort/Filter Bar	Sticky top, #0A0A0F bg, 52px height. Sort dropdown left, type pills right. Active pill: amber fill, dark text.
Search Input	320px wide, icon prefix, 36px height, #111118 bg, amber focus ring. Real-time (no submit button).

Phase Outcome: The app looks like a real file manager. Thumbnails, grid layout, lightbox, sort/filter. Every file is browsable and actionable.

PHASE 04

Polish & UX Refinement

Animations, micro-interactions, mobile layout, error handling

Duration	3–4 days
Goal	Elevate from functional to delightful. Every interaction feels smooth. Mobile works. Edge cases are handled gracefully.
Stack	CSS transitions/animations, ResizeObserver, localStorage fallback
Outcome	A polished, production-quality feel. Zero rough edges. Works on phone.

Tasks

#	Task	What to build / design	Effort
1	Page Load Animation	Staggered reveal on mount: sidebar slides in from left (150ms), header fades down (100ms), file cards fan in with 30ms delay each. CSS keyframes only.	3h
2	Folder Switch Transition	On folder select: file grid fades out (100ms) then new grid fades in (200ms). Sidebar active item slides with smooth background transition.	2h
3	Upload Micro-interaction	On drop: burst particle effect (CSS only, 6 dots scatter from drop point). Each new file card 'pops' in with scale(0.8)->scale(1) + opacity 0->1.	3h
4	Delete Animation	On confirm delete: card collapses height to 0 + opacity fades to 0 over 300ms. Grid reflows smoothly. Undo toast appears simultaneously.	2h
5	Undo Delete	10-second undo window after deletion. Amber toast with countdown progress bar. Click Undo: card re-inserts at original position with pop-in animation.	3h
6	Mobile Layout	< 768px: sidebar collapses to bottom drawer (swipe up to open). File grid goes single-column. Floating action button (+) for upload. Touch-friendly tap targets (48px min).	4h
7	Empty State Illustrations	SVG illustrations (CSS-drawn, no external images) for: no folders, empty folder, no search results. Each unique, on-theme, animated (subtle float keyframe).	3h

#	Task	What to build / design	Effort
8	Error Boundary	Wrap app in React error boundary. On crash: friendly error card with 'Reload' button. Storage quota error: specific message + tips to clear space.	2h
9	Accessibility Pass	Add ARIA roles, labels, live regions for toast notifications. Full keyboard navigation. Focus trap in modals. Colour contrast audit (WCAG AA).	3h
10	Performance Audit	Lazy-load thumbnails with IntersectionObserver. Debounce search input (150ms). Memoize filtered file list. Target < 100ms interaction latency.	2h

UI Specifications

UI Component	Design Specification
Animation Easing	All transitions use cubic-bezier(0.34, 1.56, 0.64, 1) for springy feel on entrances. Exits: linear 150ms. Never use ease-in for entrances.
Undo Toast	Fixed bottom-centre, 400px wide, purple accent, countdown bar (100%->0% over 10s). Bold 'Undo' button right-aligned. Dismissible.
Mobile Drawer	Bottom sheet, 80vh max-height, handle bar at top, spring animation (translateY 100%->0). Backdrop closes it. Border-radius 20px top corners.
Floating Action Btn	56px circle, purple fill, + icon, fixed bottom-right (16px margin). Ripple on tap. Expands to show 'Upload' label on hover (desktop).
SVG Empty States	Centred, max 200px wide, 60% opacity. Slow float animation (6s ease-in-out infinite alternate, translateY -8px). Caption below in text-muted.
Focus Rings	2px solid purple (#A855F7) with 2px offset. Applies to all interactive elements. Never remove :focus-visible outline.

Phase Outcome: The app is now genuinely delightful to use. Every interaction has feedback. Mobile users can access it. Edge cases don't crash anything.

PHASE 05

Deploy & Access

Hosting, shareable URL, optional PIN lock, cross-device testing

Duration	2–3 days
Goal	Make the app accessible from any device via a stable URL. Add optional PIN protection. Verify cross-device data sync.
Stack	Claude Artifact hosting (window.storage), optional: Vercel/Netlify for self-host fallback
Outcome	A single URL you can open on any lab machine or laptop and all data is there.

Tasks

#	Task	What to build / design	Effort
1	Artifact Deployment	Package the app as a single Claude artifact (JSX). Verify window.storage API works in artifact context. Confirm data is user-scoped (not shared globally).	2h
2	Self-Host Export	Build a standalone HTML export (single file, all JS inlined) as fallback for environments where Claude is blocked. Webpack or esbuild bundle.	3h
3	Optional PIN Lock	Settings panel: enable PIN toggle. 4-digit PIN stored hashed (SHA-256, not plaintext). Lock screen on app open if PIN enabled. 3 failed attempts: 30s lockout.	4h
4	PIN Lock UI	Full-screen lock screen: app name + icon, 4-dot PIN input (large touch targets), shake animation on wrong PIN, biometric hint text.	3h
5	Cross-Device Smoke Test	Test on: Chrome (lab PC), Firefox (lab PC), Safari (iPhone), Chrome (Android). Verify folders and files appear consistently. Document any discrepancies.	2h
6	Network Restrictions Test	Test behind university proxy/firewall. Identify if artifact URL is blocked. If so, provide self-host fallback instructions + one-click export button in settings.	2h
7	Settings Panel	Gear icon in header -> slide-in settings drawer. Options: enable PIN, export all data as ZIP, clear all data (with triple-confirm), app version info.	3h

#	Task	What to build / design	Effort
8	Data Export	Settings: 'Export all data' -> generates JSON backup of all folders + files. Download as lab-vault-backup-[date].json. Import option for restore.	2h
9	Onboarding Flow	First-visit: 3-step tooltip walkthrough (Create a folder -> Upload files -> Open anywhere). Skippable. Stored in storage so it never repeats.	2h

UI Specifications

UI Component	Design Specification
PIN Lock Screen	Full viewport, dark bg (#0A0A0F), centred card 320px wide. 4 large circles (dots) for PIN input. Numpad or keyboard input. Sky-blue accent.
Settings Drawer	Slides in from right, 320px wide, backdrop-filter blur behind. Section headers in muted caps. Toggle switches (custom CSS, sky accent). Destructive actions in red.
Export Button	Secondary style (border only, sky accent). Icon: download arrow. Progress indicator during ZIP generation. Success: brief green flash.
Onboarding Tooltip	Floating tooltip with arrow, sky accent border, step counter (1/3). 'Next' and 'Skip' buttons. Highlights the relevant UI element with ring overlay.
Lock Animation	Wrong PIN: card shakes (translateX keyframe, 6 frames, 400ms). Correct PIN: card scales up + fades out, app fades in (300ms). Smooth and satisfying.

Phase Outcome: The Lab Screenshot Manager is live, accessible, and protected. One URL, any device, always in sync. No more emailing yourself screenshots.

Phase Summary

#	Phase	Theme	Key Deliverable	Duration
01	Foundation	Design tokens, app shell, folder sidebar + creation	Design tokens, app shell, folder sidebar + creation	3–4 days
02	Upload	Drag-and-drop upload, Base64 storage, file persistence	Drag-and-drop upload, Base64 storage, file persistence	3–4 days
03	Browse	Thumbnail grid, lightbox viewer, search, sort/filter, delete	Thumbnail grid, lightbox viewer, search, sort/filter, delete	4–5 days
04	Polish	Animations, undo, mobile layout, accessibility, performance	Animations, undo, mobile layout, accessibility, performance	3–4 days
05	Deploy	Deployment, PIN lock, settings, export, cross-device test	Deployment, PIN lock, settings, export, cross-device test	2–3 days

Total: 15–20 working days (3–4 weeks) · Solo developer, focused sprints

Build tip

Complete each phase in order. Do not skip Phase 1 — the design tokens and layout decisions made there will determine the quality ceiling of every phase that follows. Polish (Phase 4) is what separates a tool people tolerate from a tool people enjoy.